

User Manual and System Administration Guide

CSCD602: Advanced Software Engineering

Capstone Project Coursework — Group Capstone Project, Individual Submission

University of Ghana, Department of Computer Science

Group Name: Architects

Project Title: HealthEasy-G HMS

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Software architecture, Ollama clinical assistant and GSTG rule engine, Next.js App Router,
PostgreSQL and Prisma, HeFRA/GHS compliance, multi-facility management

Live system: <https://healtheasy-g.vercel.app>

Repository: <https://github.com/NketiaAsubontengErnest/HealthEasy-G>

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Part I — Getting Started

1. What This System Is For

HealthEasy-G keeps one record per patient and carries it through every department of the hospital. When the front desk registers a patient, the triage nurse sees them. When the doctor orders a test, the laboratory sees the request and the cashier sees the charge. Nobody retypes a name, and nobody has to find a paper folder.

Your view of the system depends on the role your account holds. A nurse sees triage and wards; a cashier sees billing. This is deliberate — you see the work that is yours.

2. Signing In

1. Open <https://healtheasy-g.vercel.app> in a web browser.
2. Select **Staff sign in** at the top right.
3. Enter your hospital email address and password.
4. Select **Sign In to Hospital Portal**.

You will arrive at the dashboard for your role.

If sign-in fails. The message *“Invalid user credentials”* means the email address or the password is wrong. It appears for both, deliberately, so that nobody can discover which addresses are registered. If you see *“This staff account is not active”*, contact HR — your account has been suspended. After ten failed attempts the system pauses sign-in from your location for fifteen minutes.

If you are redirected to sign-in unexpectedly, your session has expired. Sessions last one eight-hour shift. Sign in again.

3. Finding Your Way Around

Area	Where	What it does
Sidebar	Left	Lists only the modules your role may open
Facility selector	Top	Shows which hospital branch you are working in

Search	Top	Finds patients by name or MRN
Role badge	Top right	Shows the role your account holds
Theme toggle	Top right (☾ / ☀)	Switches between light and dark; your choice is remembered
Notifications	Top right	Alerts needing attention
Profile	Top right	Your details and Sign out

4. Two Things That Are Not Faults

Some modules are missing from your sidebar. You only see what your role may use. If you need a module that is not there, your role does not have that authority — speak to your supervisor rather than trying another route.

A red “Access Denied” screen. If you reach a page your role may not open, the system explains which role you hold and which authority is missing. This is the system protecting patient data, not a malfunction.

Part II — Working by Role

5. OPD / Medical Records Officer

5.1 Registering a New Patient

1. Open **Patient Registration**.
2. Select **Register New Patient**.
3. Complete the form:
 - **Full name, date of birth, gender, telephone** — required.
 - **Ghana Card number** — required. This is how the system knows one patient from another.
 - **NHIS number, status and expiry** — if the patient is insured.
 - **Patient category** — Cash, NHIS, Corporate, Private Insurance or Exempted.
 - **Digital (GPS) address and residential address**.

- **Next of kin** — name, relationship and telephone.
- **Allergies and chronic conditions** — separate multiple entries with commas. Take care here: this is what warns a prescriber later.
- **Blood group.**

4. Select **Register Patient**.

The system allocates a Medical Record Number (for example HG-2026-0006) and places the patient in the triage queue automatically.

If the Ghana Card is already registered, the system refuses and tells you who holds it and under which MRN. Do not create a second record. Retrieve the existing one. Two folders for one person is the problem this system exists to prevent.

5.2 Finding an Existing Patient

Open **Patient Registration** and type a name, MRN or Ghana Card number into the search box. Results filter as you type.

5.3 Emergency Registration

For an unconscious or unidentified patient, use the fast-track option on the **Emergency** screen. It creates a record with a placeholder identifier so that care can begin at once. Complete the identification later.

5.4 Routing a Patient

From **Patient Flow & Queues**, select the patient and choose the destination department. The system issues a ticket numbered for that department — TRG-004 for triage, OPD-012 for consultation.

6. Nurse — Triage and Wards

6.1 Recording Observations

1. Open **Triage**.
2. Select the patient from the waiting list.

3. Check the banner: **allergies and chronic conditions are shown before you begin.**
4. Record blood pressure, pulse, temperature, respiratory rate, oxygen saturation, weight, height, blood glucose and pain score.
5. Choose the **ESI acuity level:**

Level	Meaning	Action
ESI-1	Resuscitation	Immediate
ESI-2	Emergency	Within minutes
ESI-3	Urgent	Standard OPD
ESI-4	Less urgent	Routine
ESI-5	Non-urgent	Routine

6. Add nursing notes.
7. Select **Save and route.**

The system computes the BMI and raises abnormality alerts itself — hypertensive crisis, low oxygen saturation, fever, tachycardia and others — against Ghana Health Service triage thresholds. You do not need to work these out; check that they match what you see in front of you.

Your name is recorded against the observations automatically, from your sign-in. You cannot record observations in someone else's name.

6.2 Ward Duties

Bed board. Open **Inpatient, Wards & Beds** to see every bed and its state: Available, Occupied, Reserved, Cleaning, Maintenance or Isolation.

Admitting. Select an available bed and assign the patient. If the bed is already occupied by someone else, the system refuses and names the current occupant.

Medication administration. The chart lists doses due. For each:

- **Administered** — records your name and the time automatically.

- **Omitted or Refused** — you must give a reason. The system will not accept a blank one, because an unexplained missed dose is the thing an audit will ask about.

7. Doctor — Consultation

7.1 Conducting a Consultation

1. Open **EMR & Consultation**. Your waiting list appears.
2. Select a patient. Before writing anything, review what is already there:
 - Demographics, **allergies** and chronic conditions
 - The triage observations just recorded, with any alerts
 - Previous encounters, with their diagnoses and plans
 - Any laboratory or imaging results
3. Record presenting complaint, history, past medical history and examination findings.
4. Assign **ICD-10 diagnoses** — one or more.
5. Raise orders as required:
 - **Laboratory** — select from the hospital test catalogue
 - **Radiology** — modality and body part
 - **Prescription** — from pharmacy stock
 - **Admission** — to a ward
 - **Procedure**
6. Record the treatment plan.
7. Issue a sick-leave certificate if needed.
8. Select **Sign and complete**.

Signing does several things at once: the note is filed under your name, each order becomes a work item in the receiving department, each chargeable item becomes a billing line, and for an insured patient an NHIS claim line is prepared from the diagnosis you coded. You do not notify the laboratory, the pharmacy or the cashier separately.

7.2 Using the Clinical Assistant

Within the prescription section, select **Suggest medicines**. The assistant reads the patient's diagnoses, observations, allergies, chronic conditions, recent results and history, and proposes medicines.

Three things to understand about it:

1. **It can only suggest what the pharmacy actually holds.** Suggestions are filtered against live stock.
2. **Allergy conflicts are flagged in red**, and that check is performed by the system itself, not taken on the assistant's word.
3. **It is advice, not a prescription.** Nothing is prescribed until you prescribe it. The decision and the responsibility remain yours.

If the assistant is unavailable, the system falls back to a rule engine based on the Ghana Standard Treatment Guidelines. The response indicates which produced it.

8. Laboratory Technician

1. Open **Laboratory**. The worklist shows orders awaiting action.
2. **Receive the specimen** — match the barcode on the container to the order.
3. Set the status to **Specimen Collected**, then **In Analysis**.
4. Enter each result parameter with its value, unit and reference range, marking abnormal or critical values.
5. **Verify** to release the result.

Verifying is a separate authority from recording. When you verify, your name and the time are stamped on the result and it becomes visible on the clinical record. If your role may record but not verify, the system will say so — a Laboratory Scientist must release it.

9. Radiographer and Radiologist

Radiographer. Open **Radiology**, select a scheduled study, confirm pregnancy screening where relevant, perform the study, upload the images and add technical notes. You cannot write the interpretation — the system will refuse, and this is correct.

Radiologist. Select a completed study, review the images, write the report, mark any critical findings and sign. Your name and the time are recorded on the report.

10. Pharmacist

10.1 Dispensing

1. Open **Pharmacy**. Prescriptions awaiting dispensing are listed.
2. Select a prescription and review it against the patient's allergies.
3. Choose a batch. **Stock is listed with the earliest expiry first** — dispense from the top unless there is a reason not to.
4. Enter the quantity.
5. Add counselling notes.
6. Select **Dispense**.

The dispense record and the stock deduction happen together. If either cannot complete, neither does — you will never have handed out medicine the ledger does not know about.

The system will refuse to dispense more than the batch holds (it tells you how many remain) or from an expired batch (it tells you the expiry date). If two counters dispense the same batch at the same instant, one is asked to retry rather than the stock going negative.

Controlled substances are marked. Handle them under the usual restrictions.

10.2 Stock

The stock list shows quantity, reorder level and expiry. Items at or below their reorder level are flagged Low Stock; items expiring within 90 days are flagged Near Expiry.

11. Cashier

1. Open **Billing & Cashier**.
2. Find the patient's invoice. Line items show what was charged, what NHIS covers and what the patient owes.
3. Take payment: cash, MTN MoMo, Telecel Cash, bank card, NHIS claim, private insurance or corporate account.
4. Enter the amount and select **Record payment**.
5. Issue the receipt.

The invoice total is calculated by the system from the line items — it cannot be edited to disagree with what was actually charged. Overpayment is refused, and the system states both figures.

Refunds. Reducing an amount already paid is a refund and requires Hospital Admin authorisation. The system will tell you this rather than allowing it silently.

12. Claims Officer

1. Open **NHIS Claims**. Claim lines generated from consultations are listed.
2. Review any **audit flags** — a missing NHIS number, an expired card, or no diagnosis code. Flagged claims are held as Draft. Resolve the flag before proceeding, because these are the conditions NHIA rejects claims for.
3. Select **Prepare batch** for the month. The system gathers every validated, unbatched claim and computes the count and total from the claims themselves.
4. Move the batch through **Exported** → **Submitted** → **Reconciled** as you deal with NHIA. Reconciling marks every claim in the batch as paid.

A claim cannot appear in two batches — the system stamps them as it batches.

13. Ward Manager, Store Keeper, Procurement, HR and Finance

Ward Manager — bed allocation, admissions, discharges and transfers, plus the ward roster.

Store Keeper — Inventory & Procurement: receive stock, record batch and expiry, transfer between stores. Status (In Stock, Low Stock, Near Expiry, Expired) is worked out by the system from quantity and expiry.

Procurement Officer — purchase requests and supplier records. Approval rests with Hospital Admin or the Director.

HR Officer — Facility Configuration: staff records and professional licences. Licences expiring within 90 days are flagged automatically. HR has no access to patient records.

Finance Officer — revenue reports and claim reconciliation. No access to clinical notes.

14. Hospital Director and Hospital Admin

Director — the dashboard gives occupancy, attendance, revenue and claim position across the facility, plus DHIMS2 returns and HeFRA licence status. The Director oversees rather than performs clinical work.

Hospital Admin — staffing, facility configuration, refund authorisation and purchase approval.

15. System Auditor

Open **Security Audit** to review the trail. Every entry records who acted, in what role, what they did, to which patient, when, and from what address. Entries cannot be edited or deleted by anyone, including the administrator.

Filter by user, action, date or patient to investigate a specific question.

Part III — System Administration

16. Administrator Responsibilities

The Super Admin manages the system, not the clinical record.

Task	Where
Register and configure facilities	Hospitals Management
HeFRA licence tracking	Facility Configuration
Staff accounts and roles	Facility Configuration
Security audit review	Security Audit
Aggregate operational statistics	Dashboard

The administrator cannot open patient records, observations or consultation notes.

This is enforced by the server, and it is intentional: section 20 of the Data Protection Act, 2012 (Act 843) limits access to personal health data to those with a care or administrative need. A technical administrator has neither. Operational oversight is provided instead by aggregate figures that contain no patient identifiers.

17. Installation

Requirements: Node.js 20 or later; PostgreSQL 14 or later.

```
git clone https://github.com/NketiaAsubontengErnest/HealthEasy-G.git
```

```
cd HealthEasy-G
```

```
cp .env.example .env
```

```
npm install
```

Edit .env:

```
DATABASE_TARGET="cloud"      # or "local"
```

```
DATABASE_URL_LOCAL="postgresql://postgres:password@localhost:5432/healtheasy_g?schema=public"
```

```
DATABASE_URL_CLOUD="postgresql://user:password@host.neon.tech/neondb?sslmode=require"
```

```
SESSION_SECRET="a-long-random-value-of-at-least-32-characters"
```

```
SESSION_MAX_AGE_SECONDS="28800"
```

Generate the session secret with:

```
node -e "console.log(require('crypto').randomBytes(48).toString('base64url'))"
```

Create and populate the database:

```
npm run db:setup:cloud    # online database
```

```
npm run db:setup:local    # offline database
```

Start:

```
npm run dev    # development
```

```
npm run build && npm start    # production
```

18. Running Online and Offline

The system runs against a local PostgreSQL when the facility has no internet, and against the managed cloud database when it has. The schema and the seed are identical.

To switch, change DATABASE_TARGET in .env to local or cloud and restart.

Command	Effect
npm run db:setup:local	Create and seed the offline database
npm run db:setup:cloud	Create and seed the online database
npm run db:studio	Open a database browser on the current target
npx tsx scripts/db-inspect.mjs	Print row counts and column types

Set both up once, and you can then work either way by changing one line.

Records created offline stay in the local database. Synchronising the two is planned but not implemented (see the Maintenance and Evolution plan).

19. Managing Staff Accounts

Accounts are created in **Facility Configuration**. Each requires a name, hospital email address, staff ID, department and role. The role determines everything the account may do.

Passwords. Set a strong initial password of at least twelve characters and require the holder to change it. Passwords are stored only as bcrypt hashes — nobody, including the administrator, can read an existing password.

Suspending an account. Change its status from Active. Sign-in is refused immediately and the attempt is recorded.

Never share accounts. The audit trail attributes every action to the account that performed it. A shared account makes the trail worthless.

20. Routine Maintenance

Task	Frequency	How
Verify backups	Daily	Confirm the provider's automated backup completed
Review audit trail	Weekly	Security Audit — look for unusual access patterns
Review licence expiry	Monthly	Facility Configuration — staff and HeFRA
Apply dependency updates	Monthly	npm outdated, then update and run npm test
Apply security patches	As released	Immediately for anything rated High or Critical
Rotate the session secret	Quarterly	Change SESSION_SECRET; all sessions end and users sign in again
Verify a restore	Quarterly	Restore a backup to a scratch database and confirm

21. Backup and Recovery

Backup. The managed database provider takes automated daily backups with point-in-time recovery. For a local database, schedule:

```
pg_dump "$DATABASE_URL_LOCAL" > healtheasy_g_$(date +%F).sql
```

Keep backups off the machine that runs the database.

Recovery.

```
psql "$DATABASE_URL_LOCAL" < healtheasy_g_2026-08-14.sql
```

```
npx tsx scripts/db-inspect.mjs # confirm row counts
```

After any restore, verify that the audit trail is intact — it is the record of what happened before the failure.

22. Troubleshooting

Symptom	Likely cause	Action
All users redirected to sign-in	SESSION_SECRET changed or unset	Restore the secret, or have users sign in again
“Cannot reach the authentication server”	Application or network down	Check the deployment; check connectivity
Screens show no data but the system responds	Role has no access to those collections	Confirm the role is correct — this may be expected
A save reports failure	Validation, authorisation or database error	The message states which; act accordingly
Sign-in refused for a valid password	Account not Active, or rate limit reached	Check status in Facility Configuration; wait fifteen minutes
Slow responses	Cold start, or database in a distant region	Expected on first request after idle; see the maintenance plan
“This staff account is not active”	Account suspended	HR reactivates it
Build fails after a schema change	Prisma client not regenerated	npx prisma generate

23. Getting Help

Report a problem with: what you were doing, what you expected, what happened, your role, and the time. The time matters — it lets the administrator find the corresponding audit entry.

Never include a password in a problem report. No member of the support team will ask for one.